Large-scale Alignment Quantum Rods Film for High Efficiency Wide Color Gamut LED Display

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***Abstract*—We explored the influence of several electrospinning parameters on the fiber formation and characterized them through the Scanning Electron Microscope (SEM). By utilizing the optimized electrospinning parameters, large-scale alignment quantum rods film had been fabricated and characterized successfully. The photoluminescence (PL) result showed that the polarization ratio is able to be achieved up to 0.31, which can be used to simplify the device structures and improve the overall efficiencies in wide color gamut LED display.**

Keywords—quantum rods, electrospinning, polarization, LED display

I. INTRODUCTION

Quantum rods (QRs), as a kind of promising one-dimensional optical nano-materials, have received great attention in recent years. This material exhibits a great range of electrical and optical properties similar to the quantum dots (QDs), which is easily tunable by controlling their size and shape [1]. These properties offer numerous applications including the fluorescent biological labels [2], luminescent solar concentrators [3] and light-emitting diode (LED) [4]. As QRs have the diameters comparable to the Bohr exciton radius and the lengths ranging from 5 to 100 nm [5], they own some extraordinary properties superior to the QDs, such as broader excitation spectrum, narrower full width half maximum (FWHM), wider color gamut [6]. Besides, the linear polarization property of their excitation light makes them more distinct from the QDs [7].

Polarization is a kind of characteristic of critical importance in many display devices, especially in the LED display, of which the backlight pass through the polarizer and the transmittance filters to get a controllable spectral output. With the QRs aligned in large scale, this novel linear polarized property can be reserved. Several approaches have been proposed for aligning nanorod or nanowires, which includes radial fluid flows [8], mechanical rubbing of spin-coated NR layers [9], stretching of polymer films [1], slow solvent evaporation at a water/air interface [10], electrically driven alignment [11], coffee stain evaporation [12], liquid crystal self-alignment [13], Langmuir-Bodgett deposition [14].

One way to fabricate the aligned composite QRs/nanofibers in large scale is the electrospinning technique, which has been confirmed the capability to achieve nanofibers with nanorods aligned over areas from nano-scale to macro-scale [15]. Usually, an electrospinning system is composed of three major parts: the high voltage power supply, the injection module and the collecting plane [16]. In the electrospinning process, the polymer solution is extruded from a needle-like injection head to form Taylor cone in the presence of electrostatic forces, and then split off into multiple fibers. By utilizing the electrospinning method, QRs can be aligned on a large-scale to maintain the macro-scale polarization in the organic polymer nanofibers.

The electrospinning parameters have a great impact on the obtained nanofibers. These parameters can be divided into three main categories: spinning solution parameters, process parameters and environmental parameters. The concentration of the polymer solution, voltage and rotation speed of the collector can directly affect the morphology, diameter and orientation degree of the fibers respectively [16]. In this study, the electrospinning parameters were optimized by a series of experiments. With the optimized parameters, an alignment QRs/polymer film was prepared successfully on a large-scale with both wide gamut and linear polarization properties.

II. EXPERIMENTAL DETAIL

We use high-quality CdSe/CdS core-shell QRs synthesized using the method described in Refs. [17]. The TEM of the QRs is shown in Fig. 1. After centrifugal purification, the QRs were mixed in the PMMA/chloroform solution with a low speed to get a well-dispersive QRs spinning solution with different concentrations. Then we investigated the influence of voltage and rotation speed on the fiber formation and alignment during the electrospinning process. The temperature was kept 318 K to ensure solvent evaporated completely. The injection speed was around 1.0 mL/h. After the electrospinning, samples were baked at 333 K for 5 hours in vacuum environment.

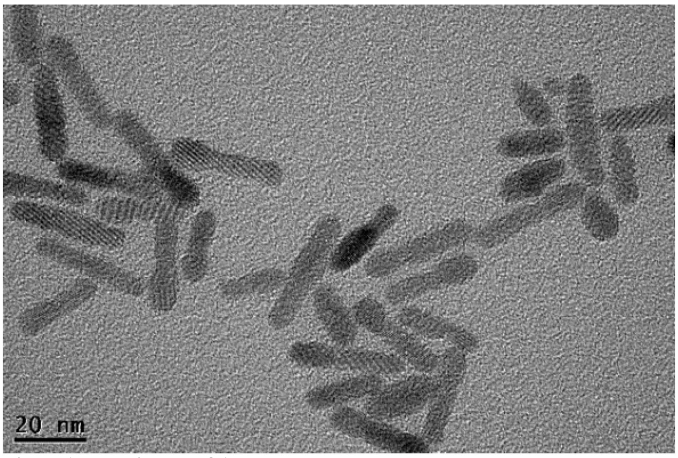

Figure 1. TEM image of the QRs

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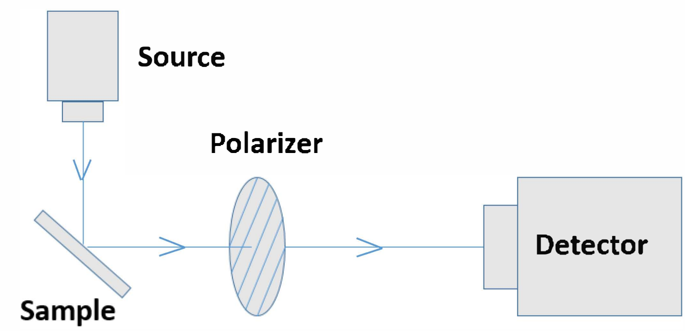

Figure 2. Schematic diagram of the polarized photoluminescence measurement device

SEM, TEM and fluorescence spectrometer were utilized to observe the morphology of the fibers, the alignment of QRs in the fibers, the photoluminescence (PL) spectrum and the polarization of the film.

Fig. 2 illustrated a simple diagram of the PL measurement process. By changing the angle of the polarizer, we got a series of the emission spectrum of the thin film. Then we took the maximum point as the representative value and got the relation of the polarizer angles and emission intensity.

III. Result and Discussion

A. Polymer Concentration

With the increase of the concentration of polymer solution, the polymer molecular chain will be entangled in the solution. The exposure concentration is defined as the concentration the polymer molecular start to tangle each other, which will change with various parameters. In stable solution, the exposure concentration is defined as c^* [18]. According to Gupta's electrospinning experiments on PMMA solution [19], the concentration is defined as c, which can influence the morphology of fiber in electrospinning process. When using the dilute solution, only beads will appear for the lack of enough tangle of polymer molecular. When $c^*/c \approx 3$, the fiber with beads will be got because polymer molecular start to tangle in fiber. If $c^*/c>3$, the fiber will be uniform and no beads will appear. Besides, the experiment on PLA [20] and PS [21] also confirm the trend, although different material may have different exposure concentration.

In our study, we prepared QRs/PMMA/chloroform solution with concentration for PMMA to chloroform of 15%, 17%, 19%, 21% and 23%. After the electrospinning, a large-scale QRs thin film was obtained. SEM was used to observe the morphology of the fibers.

Fig. 3 showed the concentration's influence on the fiber's formation. When the concentration was 15% and 17%, there were some beads appearing on the fibers shown in Fig. 3(b) and 3(d). When the concentration was 19%, 21% and 23%, there was no beads appearing on the fibers as shown in Fig. 3(f), 3(h) and 3(j).

The influence of concentration on the number of the beads also can be seen from the Fig. 3. When the concentration was 15%, the beads appear as ball shape, shown in Fig. 3(b). For

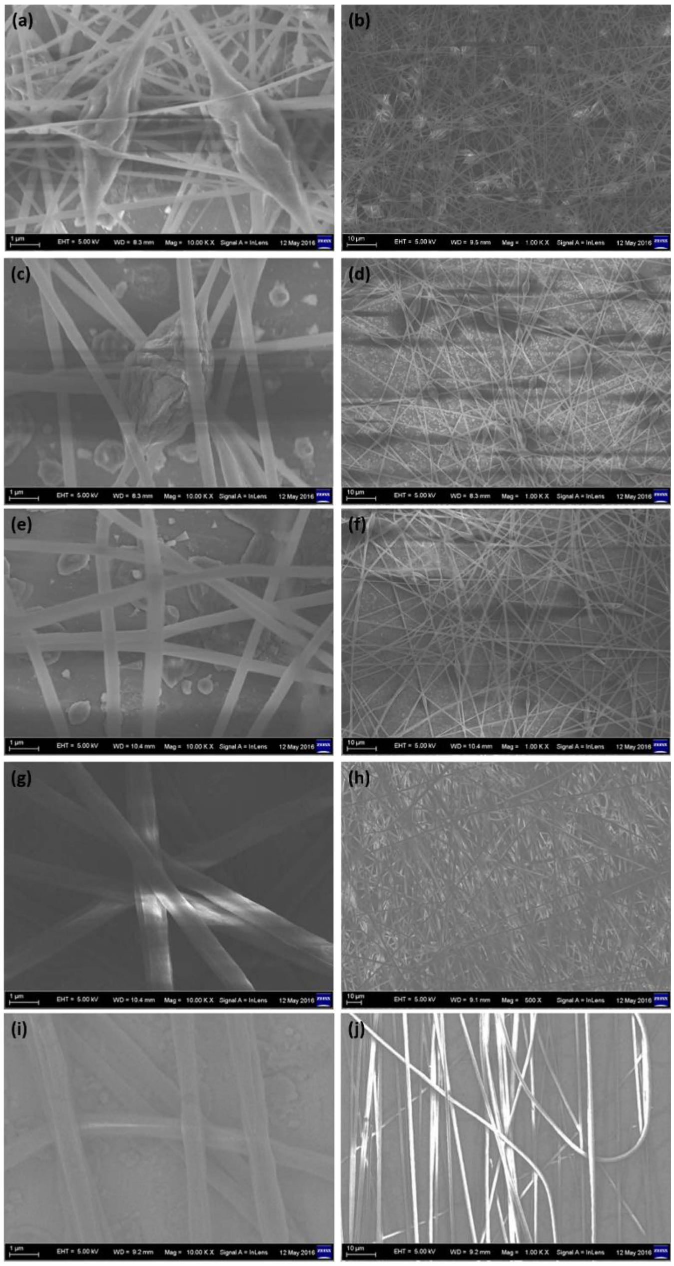

Figure 3. The SEM images with the PMMA concentration of (a)(b) 15% (c)(d) 17% (e)(f) 19% (g)(h) 21% (i)(j) 23%

the concentration of 17%, the beads appeared as fusiform shape shown in Fig. 3(d). For the concentration of 19%, there was nearly no beads on fibers, shown in the Fig. 3(f).

We also can see that the influence of the concentration on the fiber's diameter in Fig. 3(e), 3(g) and 3(i). When the concentration was 19%, the fiber's diameter was around 200nm shown in Fig. 3(e). As for the concentration of 21%, the fiber's diameter was around 400nm shown in Fig. 3(g). When the concentration was 23%, the fiber's diameters was around 500nm in Fig. 3(i). Based on these, we can draw the conclusion that the higher the concentration is, the thinner the diameter is.

The concentration above 27% was also tried but it turned out to be that 27% concentration is too high to form the fiber. The polymer was solidified at the syringe. The results showed that 19% was supposed to be the proper exposure concentration for QRs/PMMA composite material. For the concentration below 19%, beads would appear on the fibers and the number of them decreased as the concentration increased. For the concentration during 19% and 27%, there were no beads appear on the fiber but the diameter of the fiber was correlated with the concentration positively.

B. Voltage

In the electrospinning process, the micro jet produced by the electric force overcome the surface tension. There exists a critical voltage, which is the minimum voltage of producing the jet [22]. Similarly, there is an appropriate voltage for the fibers. If a voltage between the critical and suitable voltage is applied, some beads will appear on the fiber [20,23,24]. For a definite concentration, because the charge on the jet increased and the acceleration increased, the diameter of the fiber will decrease with the voltage increase, which can be described in the following equation [25]:

$$r_t = [\gamma \varepsilon \frac{Q^2}{I^2} \frac{2}{\pi(2 \ln \emptyset - 3)}]^{1/3} \qquad (1)$$

where the γ is the surface tension of the polymer solution, the ε is the dielectric coefficient, the Q is the flow rate of the polymer solution, the I is the current of the solution and the $\emptyset$ is the rate of the radius of the unstable starting point and the curvature radius of whip movement.

The r_t is the radius of the fiber, which is inverse proportional to the current I on the fiber. That means the radius will decrease as the current decreased. If the concentration is low, the fiber may break into beads for the deficiency of the entanglement between polymer molecular [26,27].

In this study, we prepared the QRs/PMMA/chloroform solution with concentration of 19%. Then we did the electrospinning under different voltages of 8 kV, 10 kV, 12 kV and 14 kV.

Some fusiform beads appeared on the fiber as the Fig. 4(a) and 4(b) shown. It showed that the voltage of 8 kV and 10 kV could not provide enough electrical field force to overcome the surface tension of polymer solution. The fusiform beads were formed along the fibers due to the surface tension. As Fig. 4(d) shown, when the applied voltage was 14 kV, some fibers were broken due to the high voltage. When the voltage was 12 kV, the fibers were obtained with no beads and no fractures as shown in Fig. 4(c). The 12 kV was supposed to be the suitable voltage for the QRs/PMMA composite fiber.

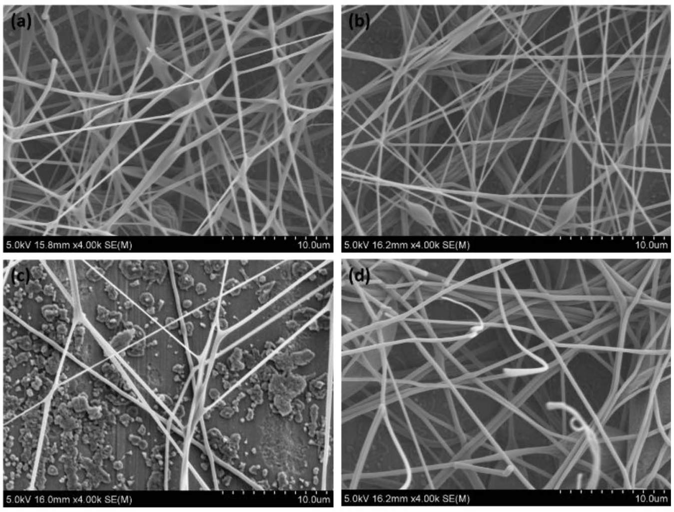

Figure 4. The SEM image with the electrospinning voltage of (a) 8 kV (b) 10 kV (c) 12 kV (d) 14 kV (Scale by 4k)

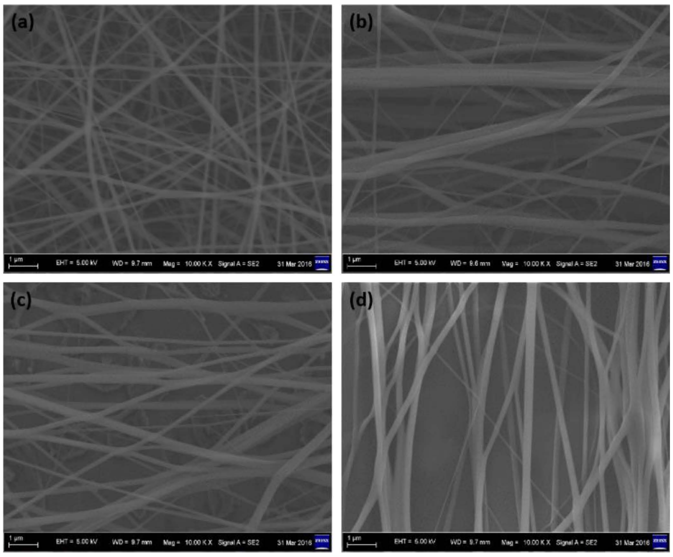

Figure 5. The SEM image with the electrospinning rotation speed of (a) 1000 rpm (b) 2000 rpm (c) 3000 rpm (d) 4000rpm (Scale by 10k)

C. Rotation Speed

In the electrospinning process, the spinning solution will split into multi-jet flow and form fibers between the sprinkler and collecting plane. Fibers will become curved due to the extension of Taylor cone [27] so that we cannot get the alignment of the nanofibers.

In this experiment, we used a high speed rotating roller as the receiving plane. The fiber would be aligned to string by the drag of the roller. We prepared the QRs/PMMA/chloroform solution with concentration of 19%. The electrospinning process was done under 12 kV with different rotation speed of 1000 rpm, 2000 rpm, 3000 rpm and 4000 rpm.

As the Fig. 5(a), 5(b), 5(c) and 5(d) shown, with the rotation speed increasing, the fibers' alignment would be better. As the higher rotation speed would lead to higher torque, the fiber would be aligned better by higher torque. However, according to our result of the rotation speed of 6000 rpm, the fibers would be flown out by the wind caused by too high rotation speed and only a little fiber could be collected on the collector, which is unavailable for our study. The 4000 rpm, a middle speed, was supposed to be the appropriate speed for QRs/PMMA composite material in electrospinning.

D. Polarization

With the optimized parameters, 21% concentration, 12 kV voltage and 4000 rpm rotation speed, a film with the scale of 30cm*5cm was obtained. That film can be used as the enhancement film in LED display for both wide gamut and

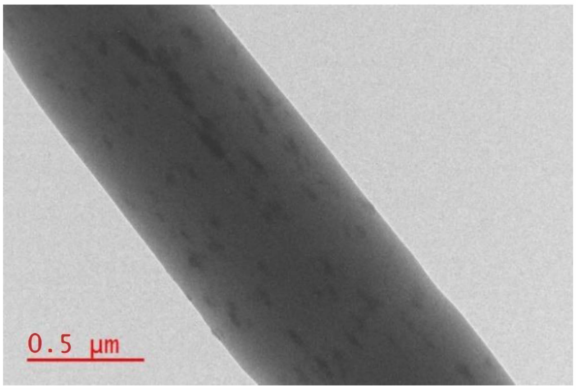

Figure 6. TEM image of the composite fiber

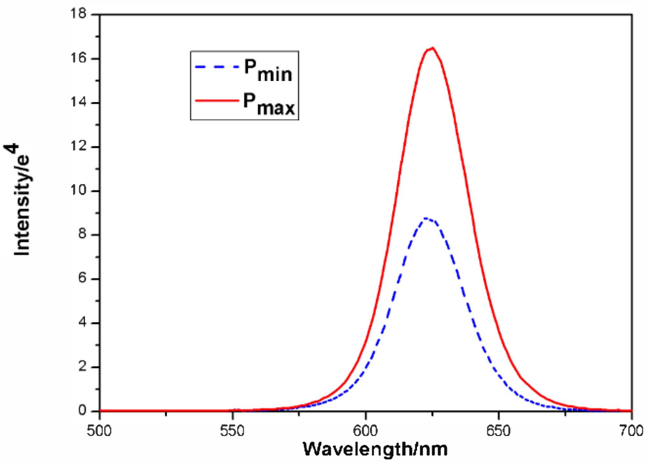

Figure 7. The PL spectral of the maximum and minimum intensity

linear polarization properties. The wide gamut is the native property for QRs, and the linear polarization property is the result of the alignment of QRs.

The whole alignment of QRs in the enhancement film is determined by the alignment of polymer fibers and the alignment of QRs in polymer fibers. TEM was used to observe the position and direction of QRs in polymer fiber. In the TEM image in Fig. 6, the QRs were generally aligned as the same direction as the fibers.

Then the polarization was measured by using the setup shown in Fig. 2. A light with 450nm wavelength was used to stimulate the aligned QRs film and the scan range was from 500nm to 700nm. P is defined as the polarization, which is calculated by the equation below:

$$P = (P_{max} - P_{min})/(P_{max} + P_{min}) \quad (2)$$

P_{max} and P_{min} are the maximum and minimum intensity, as shown in Fig. 7. The intensity peak in the PL spectrum at each angle of the polarizer was recorded and plotted as shown in Fig. 8. As we all known, the better the fiber alignment, the higher polarization. In this test, the polarization result can reach up to 0.31.

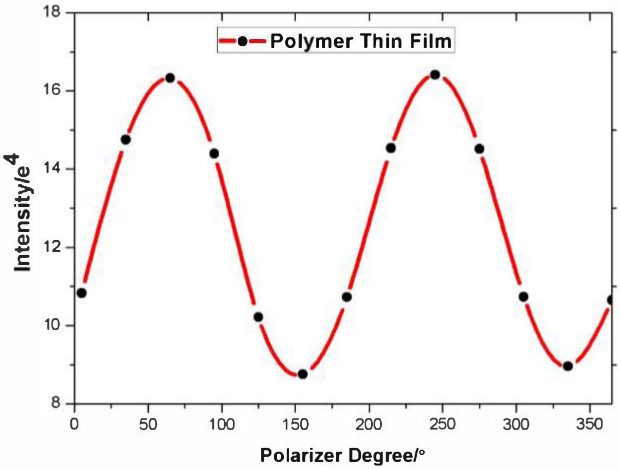

Figure 8. The polarization of the electrospinning QRs film

IV. Conclusion

In this study, we discussed the influence of the electrospinning parameters on the fiber formation. The concentration and the voltage had great impacts on the morphology and diameter of the fibers. High concentration and voltage could improve the fiber formation process and decrease the fiber diameter, leading to a better alignment of the QRs in fibers. Besides, the rotation speed of the collector also influenced the macroscopic orientation of the nanofibers. With the optimized parameters, the concentration of 19%, voltage of 12 kV and rotation speed of 4000 rpm, we obtained the aligned QRs film in large scale successfully. Moreover, we measured the PL spectrum and the results showed that the polarization of the obtained film could be achieved up to 0.31, which can be used to simplify the device structures and improve the overall efficiencies in wide color gamut LED display.

Acknowledgment

This work is supported by Climbing Program Special Funds (Grant No. pdjh2016b0445), National Natural Science Foundation of China (Grant No. 51402148), Guangdong High Tech Project (Grant No. 2014A010105005 and No. 2014TQ01C494), and Shenzhen Innovation Project (Grant No. KC2014JSQN0011A and No. JCYJ20150630145302223).

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